King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 101 Final Exam Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. $\int \left(\frac{2}{x} + \sec^2 x\right) dx =$

(a) $\frac{-2}{x^2} + \tan x + c$

(b) $2 + 2\sec x + c$

(c) $2\ln|x| + \sec x \tan x + c$

(d) $2\ln|x| + \tan x + c$

(e) $2\ln|x| + \sec x + c$

2. If a manufacturer wants to design an open box having a square base and a surface area of $192\,\mathrm{m}^2$, then the side length of the square base that will produce a box with maximum volume is

(a) 4

(b) 6

(c) 8

(d) 9

(e) 10

3. Which of the following statements is true about the graph of $f(x) = e^{-(x-3)^2}$?

(a) It is concave up on $(3, \infty)$.

(b) It is concave up on $\left(3 - \frac{1}{\sqrt{2}}, 3 + \frac{1}{\sqrt{2}}\right)$.

(c) It is concave down on $\left(-\infty, 3 - \frac{1}{\sqrt{2}}\right)$.

(d) It is concave down on $(-\infty, \infty)$.

(e) It is concave up on $\left(3 + \frac{1}{\sqrt{2}}, \infty\right)$.

4. $\lim_{\Delta x \to 0} \dfrac{(-2+\Delta x)^3 + 8}{\Delta x} =$

(a) -2

(b) -8

(c) 8

(d) -12

(e) 12

5. $\lim_{x \to \infty} \left(1 - \dfrac{1}{x}\right)^x =$

(a) $\dfrac{1}{e}$

(b) 0

(c) ∞

(d) 1

(e) e

6. The limit $\lim_{x \to \infty} \left(\sqrt{4x^2 + x} - 2x\right) =$

(a) $-\infty$

(b) ∞

(c) $\dfrac{1}{2}$

(d) 2

(e) $\dfrac{1}{4}$

7. Let $\Delta x = 0.1$, the differential of $f(x) = \dfrac{1}{x}$ at $x = -1$ is

(a) $-\dfrac{1}{11}$

(b) $-\dfrac{1}{9}$

(c) $\dfrac{1}{10}$

(d) $\dfrac{1}{9}$

(e) $-\dfrac{1}{10}$

8. If $y = -2x + 3$ is the tangent line to the curve of $f(x) = kx^2$, then $k =$

(a) $-\frac{1}{3}$

(b) $-\frac{1}{2}$

(c) 0

(d) -2

(e) -3

9. $\int \frac{x+1}{\sqrt{x}}\,dx =$

(a) $x^{\frac{3}{2}} + \sqrt{x} + c$

(b) $\frac{2}{3}x^{3/2} + 2\sqrt{x} + c$

(c) $\frac{1}{2}(x+1)^2 + 2\sqrt{x} + c$

(d) $2(x+1)^2\sqrt{x} + c$

10. If $y = x^{\ln x}$, then $y'(e) =$

(a) e

(b) 0

(c) 1

(d) $\frac{1}{e}$

(e) 2

11. $\frac{d^2}{dx^2}\left(2\cosh(\ln x)\right) =$

(a) $1 + \frac{2}{x^3}$

(b) $\frac{2}{x^3}$

(c) $1 - \frac{1}{x^3}$

(d) $1 - \frac{2}{x^3}$

(e) $1 + \frac{1}{x^2}$

12. The number of inflection points the function $f(x) = \sin x$ has in the interval $\left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is

(a) 4

(b) 5

(c) 3

(d) 1

(e) 2

13. $\dfrac{\cosh^2 0.1 - \sinh^2 0.1}{\tanh^2 0.2 + \operatorname{sech}^2 0.2} =$

(a) 2

(b) 0.5

(c) -0.5

(d) -1

(e) 1

14. Using differential, the value of $\sqrt[3]{999.4}$ approximately equals

(a) 9.998

(b) 9.94

(c) 9.996

(d) 9.98

(e) 9.994

15. The tangent line approximation of the function $f(x) = \dfrac{1}{1+e^{-x}}$ at $x = 0$ is

(a) $y = \frac{1}{2}x + \frac{1}{4}$

(b) $y = \frac{1}{2}x - \frac{1}{4}$

(c) $y = \frac{1}{4}x + \frac{1}{2}$

(d) $y = \frac{1}{4}x$

(e) $y = \frac{1}{4}x - \frac{1}{2}$

16. Let b and c be positive numbers, and $f(x) = \dfrac{x^2 + bx + c}{x + b}$. Then

(a) $y = -b$ is a vertical asymptote.

(b) $y = x$ is a slant asymptote.

(c) f has one horizontal asymptote.

(d) f has two horizontal asymptotes.

(e) $y = \dfrac{c}{b}$ is an x-intercept.

17. $\lim_{x\to 0} \dfrac{4x - 2\sin 2x}{2x^3} =$

(a) $\dfrac{4}{3}$

(b) ∞

(c) $\dfrac{2}{3}$

(d) $\dfrac{1}{2}$

(e) 0

18. The equation of the tangent line to the curve $9x^2 + 4y^2 = 13$ at the point $(1, 1)$ is given by:

(a) $9x + 4y = 13$

(b) $9x + 4y = 8$

(c) $9x + 4y = 11$

(d) $9x - 4y = 7$

(e) $9x - 4y = 9$

19. A point that lies on the tangent line to the curve $y = \dfrac{1}{1 - x}$ at the point $\left(3, -\dfrac{1}{2}\right)$ is

(a) $(1, 1)$

(b) $(-1, 1)$

(c) $(2, 0)$

(d) $(5, 0)$

(e) $(-3, 2)$

20. A man 2 meters tall walks at a rate of 1.5 meters per second away from a light that is 5 meters above the ground. The rate at which the length of his shadow is changing when he is 3 meters from the base of the light is equal to

(a) 0.5 m/sec

(b) 1 m/sec

(c) 1.5 m/sec

(d) 2 m/sec

(e) 2.5 m/sec

21. If we have two positive numbers such that the sum of the first number squared and the second number is 108 and the product of the two numbers is a maximum, then their sum is equal to

(a) 36

(b) 66

(c) 52

(d) 88

(e) 78

22. Which of the following statements is true about the function

$$f(x) = x - \arcsin x$$

(a) f has a relative minimum at 0

(b) f is increasing on $(-1, 1)$

(c) f is increasing on $(-1, 0)$ and decreasing on $(0, 1)$

(d) f is decreasing on $(-1, 1)$

(e) f has a relative maximum at 0

23. The curve $f(x) = \dfrac{x-1}{e^x}$ intersects its horizontal asymptote at $x =$

(a) 1

(b) e

(c) e^{-2}

(d) 0

(e) e^{-1}

24. A point on the graph of $y = 4 - x^2$ that is closest to the point $(0, 0)$ is

(a) $\left(\sqrt{\frac{3}{2}}, \frac{5}{2}\right)$

(b) $(1, 3)$

(c) $\left(\sqrt{\frac{1}{2}}, \frac{7}{2}\right)$

(d) $\left(\sqrt{\frac{7}{2}}, \frac{1}{2}\right)$

(e) $(0, 4)$

25. If f is a function such that

$$f''(x) = 2, \quad f'(2) = 5, \quad f(2) = 10$$

then $f(1) =$

(a) 15

(b) 6

(c) 8

(d) 7

(e) 17

26. The number of the relative extrema of the function $f(x) = -3x^5 + 5x^3$ is

(a) 4

(b) 1

(c) 0

(d) 3

(e) 2

27. The sum of all values of c that satisfy the Mean Value Theorem for $f(x) = \dfrac{x}{x-5}$ on $[1, 4]$ is

(a) 7

(b) 5

(c) 3

(d) 9

(e) 10

28. The number x at which the function $f(x) = 2x + \dfrac{1}{x}$ has a relative minimum is

(a) $\sqrt{2}$

(b) 0

(c) $-\dfrac{1}{\sqrt{2}}$

(d) $\dfrac{1}{\sqrt{2}}$

(e) $-\sqrt{2}$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(d)**	$2\ln\lvert x\rvert + \tan x + c$.
2	**(c)**	$h = \frac{192-x^2}{4x}$, $V = \frac{192x-x^3}{4}$, $V' = 0$ gives $x = 8$.
3	**(e)**	$f'' = 2[2(x-3)^2 - 1]e^{-(x-3)^2} > 0$ for $x > 3 + \frac{1}{\sqrt{2}}$ (and $x < 3 - \frac{1}{\sqrt{2}}$).
4	**(e)**	Derivative of x^3 at -2: $3(-2)^2 = 12$.
5	**(a)**	$\ln y = \frac{\ln(1-1/x)}{1/x} \to -1$ by L'Hopital, so the limit is e^{-1}.
6	**(e)**	Conjugate: $\frac{x}{\sqrt{4x^2+x}+2x} \to \frac{1}{2+2} = \frac{1}{4}$.
7	**(e)**	$dy = -\frac{1}{x^2}dx = -1 \cdot 0.1 = -\frac{1}{10}$.
8	**(a)**	$2ka = -2$ and $ka^2 = -2a + 3$ give $a = 3$, $k = -\frac{1}{3}$.
9	**(b)**	$\int (x^{1/2} + x^{-1/2})dx = \frac{2}{3}x^{3/2} + 2\sqrt{x} + c$.
10	**(e)**	$\ln y = (\ln x)^2$, $y' = y \cdot \frac{2\ln x}{x}$, so $y'(e) = e \cdot \frac{2}{e} = 2$.
11	**(b)**	$2\cosh(\ln x) = x + \frac{1}{x}$, second derivative $\frac{2}{x^3}$.
12	**(b)**	$f'' = -\sin x$ changes sign at $x = n\pi$: $-2\pi, -\pi, 0, \pi, 2\pi$, so 5.
13	**(e)**	Both numerator and denominator equal 1, so the value is 1.
14	**(a)**	$10 + \frac{1}{300}(-0.6) = 9.998$.

Q	Ans	Brief explanation
15	**(c)**	$f(0) = \frac{1}{2}$, $f'(0) = \frac{1}{4}$: $y = \frac{1}{4}x + \frac{1}{2}$.
16	**(b)**	$f(x) = x + \frac{c}{x+b}$, so $y = x$ is a slant asymptote.
17	**(a)**	L'Hopital three times: $\frac{16\cos 2x}{12} \to \frac{4}{3}$.
18	**(a)**	$y' = -\frac{9x}{4y} = -\frac{9}{4}$ at $(1,1)$, so $9x + 4y = 13$.
19	**(d)**	$y'(3) = \frac{1}{4}$, line $y + \frac{1}{2} = \frac{1}{4}(x-3)$ contains $(5,0)$.
20	**(b)**	$s = \frac{2x}{3}$, so $\frac{ds}{dt} = \frac{2}{3}(1.5) = 1$ m/sec.
21	**(e)**	$P = 108a - a^3$, $a = 6$, $b = 72$, sum $= 78$.
22	**(d)**	$f' = 1 - \frac{1}{\sqrt{1-x^2}} \le 0$ on $(-1,1)$, so f is decreasing.
23	**(a)**	Horizontal asymptote $y = 0$; $\frac{x-1}{e^x} = 0$ gives $x = 1$.
24	**(d)**	Minimize $u + (4-u)^2$ with $u = x^2$: $u = \frac{7}{2}$, $y = \frac{1}{2}$.
25	**(b)**	$f' = 2x + 1$, $f = x^2 + x + 4$, so $f(1) = 6$.
26	**(e)**	$f' = 15x^2(1-x^2)$ changes sign only at ± 1, so 2 extrema.
27	**(c)**	$f'(c) = -\frac{5}{4}$ gives $c = 3$ or 7; only $c = 3 \in (1,4)$, sum $= 3$.
28	**(d)**	$f' = 2 - \frac{1}{x^2} = 0$, $x = \pm\frac{1}{\sqrt{2}}$; $f'' = \frac{2}{x^3} > 0$ only at $+\frac{1}{\sqrt{2}}$.